

CHAING RAI ARGOMET, Thailand

WMO: 483040

Lat: 19.8719N

Lon: 99.7793E

Elev: 401

StdP: 96.60

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.9	(c) 12.7	(d) 8.9	(e) 7.4	(f) 15.9	(g) 10.7	(h) 8.4	(i) 17.3	(j) 4.4	(k) 17.5	(l) 3.4	(m) 21.8	(n) 0.1	(o) N/A	(p) 0.257

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.8	(c) 36.7	(d) 24.3	(e) 35.3	(f) 24.4	(g) 34.1	(h) 24.4	(i) 27.2	(j) 31.7	(k) 26.7	(l) 31.3	(m) 26.4	(n) 30.9	(o) 1.0	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.1	(b) 22.5	(c) 29.7	(d) 25.6	(e) 21.9	(f) 29.2	(g) 25.2	(h) 21.4	(i) 28.7	(j) 88.9	(k) 32.0	(l) 86.9	(m) 31.4	(n) 85.3	(o) 31.0	(p) 29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 3.5	(b) 2.8	(c) 2.3	(e) 8.1	(f) 38.6	(g) 2.0	(h) 1.7	(i) 6.7	(j) 39.8	(k) 5.5	(l) 40.8	(m) 4.4	(n) 41.7	(o) 2.9	(p) 42.9		
			DB													
			WB													
				8.2	28.2	2.2	0.5	6.6	28.5	5.3	28.8	4.1	29.0	2.5	29.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	21.0	22.9	25.1	27.6	28.3	28.1	27.5	27.1	27.1	26.1	24.2	21.3		
	DBStd	3.04	2.09	1.72	1.95	2.01	1.88	1.21	1.36	1.11	1.08	1.39	1.87	2.41		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	13	6	1	1	0	0	0	0	0	0	0	0	7		
	CDD10.0	5668	341	361	469	528	566	543	541	530	513	500	427	349		
	CDD18.3	2640	88	128	212	278	308	293	283	272	263	242	177	98		
	CDH23.3	23679	583	1176	2036	2912	3159	2979	2631	2316	2259	1839	1208	581		
	CDH26.7	9354	118	474	1017	1554	1503	1167	899	707	762	635	395	123		
(14) Wind	WSAvg	0.3	0.2	0.2	0.2	0.3	0.4	0.4	0.3	0.2	0.2	0.2	0.2	0.2		
(15) Precipitation	PrecAvg	1762	13	5	27	72	202	228	296	430	258	141	60	30		
	PrecMax	2208	79	24	116	238	338	421	486	639	451	280	171	159		
	PrecMin	1235	0	0	0	2	64	105	152	259	79	29	5	0		
	PrecStd	299	23	8	36	61	82	107	87	105	101	80	48	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	33.7	36.1	38.7	39.5	35.7	35.1	33.2	33.5	33.0	32.4	30.4		
		MCWB	21.1	20.6	22.0	23.8	24.8	26.0	25.5	26.3	26.4	25.4	24.8	22.7		
	2%	DB	29.1	32.4	34.8	37.1	36.7	34.1	33.4	32.1	32.5	32.1	31.2	29.2		
		MCWB	20.8	20.8	22.1	23.9	25.2	25.8	25.8	26.2	26.2	25.2	24.0	22.1		
	5%	DB	28.1	31.2	33.6	35.7	34.9	32.8	32.1	31.2	31.6	31.2	30.3	28.1		
		MCWB	20.6	20.7	21.9	24.2	25.5	25.9	25.9	26.0	25.9	24.9	23.6	21.5		
	10%	DB	26.9	29.9	32.3	34.2	33.2	31.8	30.9	30.2	30.6	30.2	29.1	26.8		
		MCWB	20.2	20.4	21.9	24.2	25.5	25.7	25.8	25.8	25.6	24.7	23.2	20.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.5	24.0	25.3	27.0	27.6	27.5	27.5	27.2	27.5	26.5	25.6	24.1		
		MCDB	27.7	29.7	31.2	32.4	32.9	32.1	31.3	31.0	31.6	30.8	30.6	28.6		
	2%	WB	22.2	23.2	24.4	26.1	26.8	26.8	26.7	26.7	26.7	25.9	25.0	23.1		
		MCDB	26.6	29.0	30.9	32.7	32.6	31.5	31.0	30.5	31.1	30.3	29.6	27.2		
	5%	WB	21.5	22.2	23.6	25.5	26.2	26.4	26.4	26.3	26.3	25.5	24.5	22.4		
		MCDB	25.8	27.8	30.2	32.3	32.0	31.0	30.4	30.0	30.5	29.8	28.8	26.3		
	10%	WB	20.8	21.4	22.9	24.9	25.8	26.0	26.0	26.0	25.9	25.1	23.8	21.6		
		MCDB	25.2	27.0	29.2	31.7	31.3	30.6	29.8	29.6	29.8	29.2	27.8	25.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	16.1	15.3	13.1	10.0	7.8	6.8	6.6	7.4	8.4	10.5	11.6		
		MCDBR	14.4	17.5	17.3	15.6	12.7	9.5	8.8	8.1	8.9	10.0	11.4	12.8		
		MCWBR	6.6	6.9	6.0	4.7	3.8	3.0	2.9	3.1	3.5	3.9	4.8	5.9		
	5% WB	MCDBR	11.9	14.9	14.8	13.3	10.7	8.4	7.6	7.4	8.3	9.0	9.9	10.8		
		MCWBR	5.8	6.2	6.1	5.2	4.1	3.1	3.1	3.2	3.6	3.8	4.4	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.418	0.499	0.697	0.665	0.489	0.437	0.431	0.453	0.470	0.493	0.417	0.412		
	taud		2.238	2.011	1.573	1.696	2.210	2.389	2.396	2.315	2.240	2.138	2.307	2.280		
	Ebn at Noon		852	799	663	693	818	854	860	847	831	795	847	844		
	Edh at Noon		130	172	277	247	146	121	121	131	140	150	121	121		
(44) All-Sky Solar Radiation	RadAvg		4.49	5.36	5.52	5.94	5.37	4.73	4.29	4.30	4.53	4.40	4.48	4.15		
	RadStd		0.25	0.26	0.31	0.36	0.48	0.32	0.32	0.27	0.20	0.21	0.33	0.29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	+0.17	N/A	N/A	+149

Nomenclature: See separate page